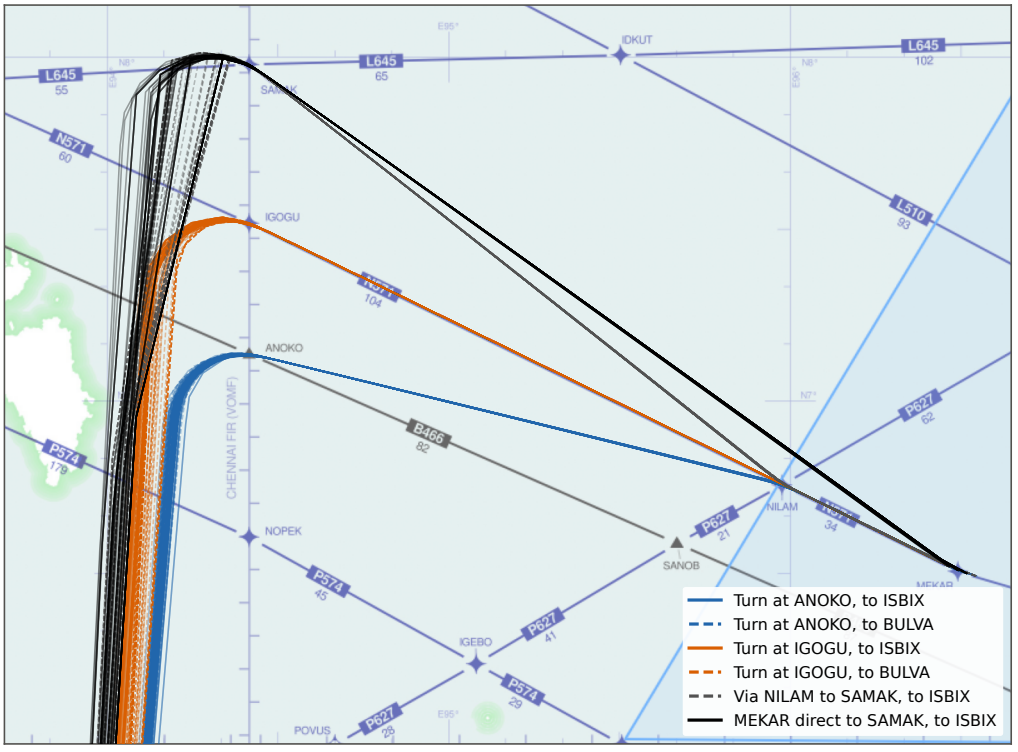
Where the paths turn south, on the en-route chart

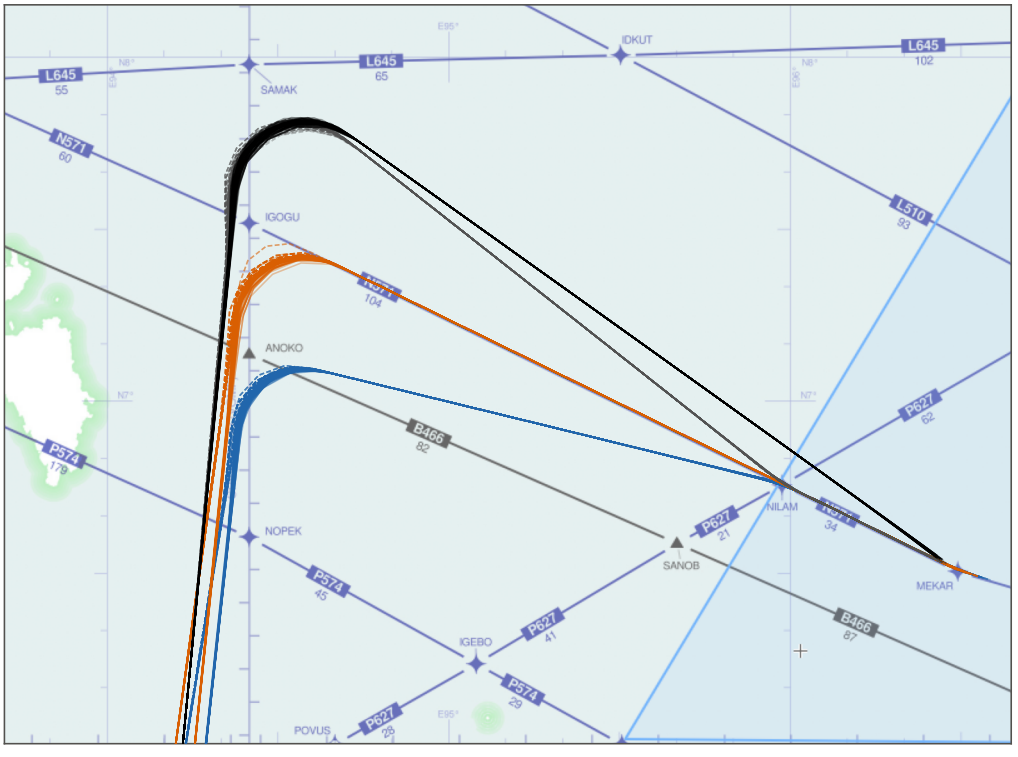
a

Fly-over turns: each path passes over the fix, then turns



b

Fly-by turns: each path starts its turn before the fix



Plain version: The lines show the flight paths that the satellite data and the radar support for each of six routes, every 40 seconds from 18:22 to 18:43. In panel a, the aircraft flies over each point and then turns. It then goes 14 to 18 NM west of the FIR boundary before it flies south. In panel b, the aircraft starts each turn before the point, as an autopilot usually does. The paths to SAMAK then turn about 18 NM before SAMAK and come south 3 to 4 NM from IGOGU. In both panels, all routes start the turn south at almost the same time, about 18:38. The chart is in the background to show the airways and the FIR boundary.

Technical version: re-flights of the run G (fly-over, turn_backs) and run H (fly-by + leg tracking) route configurations, one route per run, seed 1, 84k particles, full evidence stack as runs G/H; binary af8417eb plus the route-sampler code (off here; committed as 6d9f4ef7). 20 of 300 posterior route samples per route from output.trace (40 s, 18:22:00-18:42:40), extended with the 600 s route points to 18:51:49. Turn start = track 0.5 NM off the inbound leg (medians): fly-over 0.8-1.4 NM after the fix, 18:38:20-18:42:10; fly-by 8.9-10.1 NM before ANOKO, 12.6-13.5 NM before IGOGU, 18.0-18.2 NM before SAMAK, 18:37:32-18:37:54. Closest approach to the turn fix: fly-over 0.2-0.3 NM; fly-by ANOKO 4.9-5.8, IGOGU 7.9-8.6, SAMAK 12.2 NM. Largest distance west of 094°25'E by 18:42:40: fly-over 13.9-17.7 NM (NILAM-SAMAK 4.5), fly-by 3.7-5.2 NM. The fly-by SAMAK lead (18 NM) is shorter than the 23 NM reported earlier from a re-flight that used data only to 19:41. Background: chart image supplied by Pete (current cycle; it shows EMRAN and L510, absent from the 2013.10 data), georeferenced by an affine fit to 12 fixes, residual 1.4 px (0.7 NM) or less; north is up. Fix positions as used in the runs. Free flight is not shown.